Lab: Multi-Agent Active Inference and Shared Generative Models

> **Learning Goal:** Analyze multi-agent scenarios using Active Inference, exploring coordination, communication, and collective intelligence.

Part 1: Recursive Modeling Depth

Exercise: Consider a conversation between two people (A and B) discussing a controversial topic:

ToM Level	A's Model	Effect on Behavior
Level 0	A has beliefs about the topic	A states opinions directly
Level 1	A models B's beliefs about the topic	A adjusts message based on B's likely knowledge
Level 2	A models B's model of A's beliefs	A anticipates B's response to A's statement, and pre-addresses objections
Level 3	A models B's model of A's model of B	A considers how B will react to A's anticipation of B's objection

1. At what level does the recursion become practically useful vs. computationally intractable?
2. How does autism (proposed as reduced ToM depth) affect conversational dynamics?
3. How does Machiavellianism (proposed as deep ToM used for self-serving purposes) use deep recursion?

Write your response here

Part 2: Game Theory as Active Inference

Learning Goal: Recast classical games in Active Inference terms.

Exercise: Analyze the Prisoner's Dilemma using Active Inference:

Two agents each choose: Cooperate (C) or Defect (D). Payoffs: CC=(3,3), CD=(0,5), DC=(5,0), DD=(1,1).

Standard game theory: Nash equilibrium = (D, D) — both defect.

Active Inference reformulation:

1. Each agent has a generative model of the other agent's likely action
2. Expected free energy for Cooperate = G(C) depends on the predicted probability that the other cooperates
3. Expected free energy for Defect = G(D) depends on the same prediction

If Agent A believes (with high precision) that B will cooperate → A's best policy is Defect (exploiting trust) If Agent A is uncertain → EFE may favor Cooperate (epistemic value of maintaining relationship)

Key question: Under what conditions does cooperation emerge?

- When agents have repeated interactions (iterated PD) — long-term model updating favors cooperation
- When agents' C vectors include social preferences (prosocial preferences)
- When reputation mechanisms provide additional observations about trustworthiness

Write your response here

Part 3: Shared Model Formation

> **Learning Goal:** Trace how shared generative models form through communication.

Exercise: Model the formation of a scientific consensus:

1. **Initial state**: 20 scientists, each with their own generative model of a phenomenon. Models differ significantly.
2. **Communication**: Scientists publish papers (signals encoding their beliefs + evidence)
3. **Model updating**: Each scientist updates their model based on received papers (weighted by precision of the evidence and credibility of the source)
4. **Convergence**: Over multiple rounds, models converge → shared generative model emerges

Map this process:

Round	Communication Event	Effect on Model Diversity	Collective Free Energy
1	Initial hypotheses published	High diversity, high collective F	Maximum (conflicting models)
2	Experimental results shared	Diversity decreases for well-supported views	Decreasing
3	Review papers synthesize	Further convergence, outliers identified	Declining
N	Consensus forms	Low diversity on core claims, ongoing debate at frontiers	Near minimum

What happens when evidence dramatically contradicts the consensus? (Paradigm shift = collective model revision)

Write your response here

Part 4: Stigmergic Communication

> **Learning Goal:** Analyze indirect communication through environment modification.

Exercise: Model an ant colony foraging task using stigmergy:

1. **Individual agent**: Each ant minimizes free energy given local sensory information (pheromone concentration, food detection)
2. **Active states**: Lay pheromone (strength proportional to food quality), move in direction of pheromone gradient

3. **Collective behavior**: Pheromone trails form between nest and food sources. Shorter paths accumulate more pheromone (faster round-trip). Longer paths lose pheromone to evaporation.
4. **Emergent optimization**: The colony converges on the shortest path — without any ant knowing the global map.

How does this implement: (a) collective inference about food locations, (b) model updating through environmental modification, (c) Bayesian model selection (short path wins)?

Write your response here

Part 5: Reflection

In 300 words, reflect: Multi-agent Active Inference treats human communication as mutual model alignment. But is genuine understanding more than model alignment? When two people deeply understand each other, is something happening beyond having similar generative models?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	Recursive analysis	Theory of Mind levels
2	Game-theoretic reasoning	EFE in strategic interactions
3	Process tracing	Shared model formation
4	Collective behavior	Stigmergic communication
5	Philosophical reflection	Understanding vs. alignment